Empirical Methods in Finance - Group Project 2

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1) Interpretation of Each Retained Principal Component: We use January 2010 through December 2025 as our training window, 16 years and 192 monthly observations per stock. We start in 2010 to exclude the 2008-2009 financial crisis, which would dominate the covariance structure and distort the factors. Extending through 2025 captures the full AI investment cycle and the evolving rate environment. The window covers four meaningfully different regimes: the post-2010 bull market, the 2020 COVID shock and recovery, the 2022 rate hiking cycle, and the 2023-2025 AI-driven growth period. We filter to market cap above $2B as of December 2024 to keep liquid, investable names with full return histories.

We retain four components. The scree plot shows PC1 at 27.10%, then a sharp drop to PC2 at 3.48%, PC3 at 2.98%, and PC4 at 2.79%. From PC5 onward nothing clears 2% and the components lose economic interpretability. We retain four because each of PC2 through PC4 maps to a recognizable economic dimension.

PC1 (Broad Market / Industrial-Financial Core - 27.10%): Loads positively across almost all stocks. High loadings: PH (0.0404), AMP (0.0399), PRU (0.0394), ITW (0.0393), PFG (0.0386), industrials and diversified financials that are sensitive to broad economic cycles. Near-zero loadings include SLNO, ZM, SMMT, ZTO, BNTX, niche biotech and low-liquidity names. PC1 is the overall market factor.

PC2 (Defensive Utilities vs Regional Banks/Energy - 3.48%): High loadings: CMS (0.0942), WEC (0.0936), AWK (0.0925), XEL (0.0865), ED (0.0839), regulated utilities with stable bond-like cash flows. Negative loadings: WTFC (-0.0555), CFG (-0.0552), TCBI (-0.0543), COP (-0.0534), HP (-0.0532), regional banks and energy names that benefit from higher rates and activity. PC2 captures defensive income versus cyclical financial and energy exposure.

PC3 (Growth Tech / AI vs Defensive Staples - 2.98%): High loadings: DDOG (0.0854), NET (0.0842), ZS (0.0827), ESTC (0.0793), CRWD (0.0761), cloud, cybersecurity, and SaaS names driven by AI adoption. Negative loadings: AVA (-0.0606), NWE (-0.0556), CPB (-0.0554), ED (-0.0547), ALL (-0.0545), utilities and consumer staples with low-growth defensive cash flows. Including 2025 data brings the AI cycle into the factor structure. PC3 captures the return separation between high-growth digital businesses and low-growth defensive names.

PC4 (Energy/Real Assets vs Industrials - 2.79%): High loadings: OVV (0.0793), VTR (0.0724), WELL (0.0724), BRX (0.0713), MGY (0.0711), energy producers and REITs. Negative loadings: CNH (-0.0656), GGG (-0.0642), ODFL (-0.0564), INFA (-0.0530), QGEN (-0.0529), industrial equipment and logistics names. PC4 captures real asset and energy exposure versus industrial services.

2) View on Each Factor - Direction and Economic Reasoning

PC1 - Slight positive (weight 0.3): Broad market beta contributes positive return over any horizon but over one week it is largely noise. We keep a small weight but do not rely on it.

PC2 - Neutral, excluded: Utilities move on rate data, banks move on credit news. Too much binary risk in a single week on both sides. Excluded.

PC3 - Strong positive (weight 0.7): This is the primary bet. The AI investment cycle is the dominant market theme right now. Hyperscalers are reaffirming capex, enterprise security and cloud spend is accelerating, and names like DDOG, NET, ZS, CRWD, and ESTC move on AI news flow in the short term. On the geopolitical side, US-China trade tensions and export controls hurt hardware-exposed names. Cloud and SaaS businesses have minimal direct exposure to these restrictions. Defensive staples on the negative side of PC3 have no near-term upside catalyst. Over one week, thematic momentum matters more than long-run factor exposure, and PC3 is where we have the clearest conviction.

PC4 - Neutral, excluded: Mixed signals on commodity prices and industrial demand. Cannot call the direction over one week. Excluded.

Scoring formula: score = 0.3 x PC1 + 0.7 x PC3. Short horizon means we underweight broad market beta and overweight the AI thematic factor where we have conviction.

3) Why the Selected Stocks Match Our View

The formula selects for high PC3 loadings, producing a portfolio of cloud infrastructure, cybersecurity, fintech, and AI-adjacent software. DDOG (PC3 0.0854, score 0.0636) is the leading observability platform, a direct beneficiary of AI workload scaling. NET (PC3 0.0842, score 0.0616) provides cloud networking that benefits from rising AI traffic. ZS (PC3 0.0827, score 0.0603) is the leading cloud security platform. ESTC (PC3 0.0793, score 0.0595) provides the search and data infrastructure AI applications run on. CRWD (PC3 0.0761, score 0.0560) benefits from growing enterprise AI security budgets. Names like SQ, COIN, MRVL, PYPL, and KEYS contribute through a mix of PC1 and PC3, adding fintech and semiconductor exposure without diluting the core thesis.

4) Sector Composition and Whether It Was Intentional

Around 80% of the 30 names are cloud software, cybersecurity, or fintech. This came from the scoring formula, not manual selection. PC3 captures the AI growth tech versus defensive axis, so the highest-scoring names naturally cluster in SaaS and cloud infrastructure. We are comfortable with this given the one-week horizon and the AI investment backdrop. The main risk is a sharp risk-off move from a hawkish Fed statement, a trade escalation, or a broad selloff in high-multiple growth names, which would hit most of the portfolio at once. That is the trade-off of a concentrated thematic factor view over a short window.

Portfolio Tickers: DDOG, NET, ZS, ESTC, HUBS, CVNA, SHOP, TWLO, CRWD, AFRM, DOCU, OKTA, SQ, TWST, MDB, SPOT, TTD, TEAM, ASAN, CRSP, RNG, COIN, NOVT, W, MRVL, DASH, KEYS, PYPL, BILL, ZG